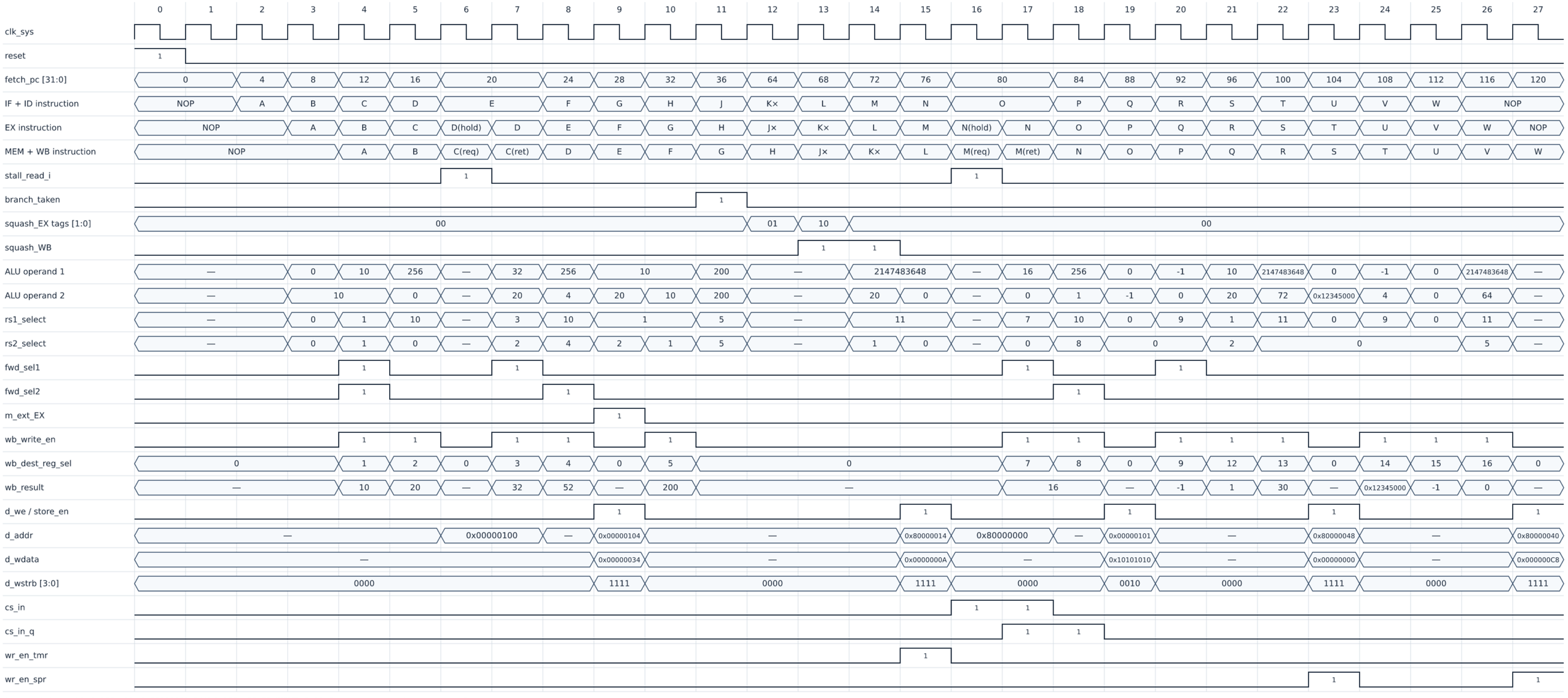
# A. RV32IM pipeline — 28 continuous clk_sys cycles

Reset, forwarding, read wait, taken / not-taken branch, squash, MUL, MMIO and byte store. Selected control assumptions: see README.

Column = after left rising edge; outputs latch at next edge. — = not consumed. // = omitted time / independent boundary window.



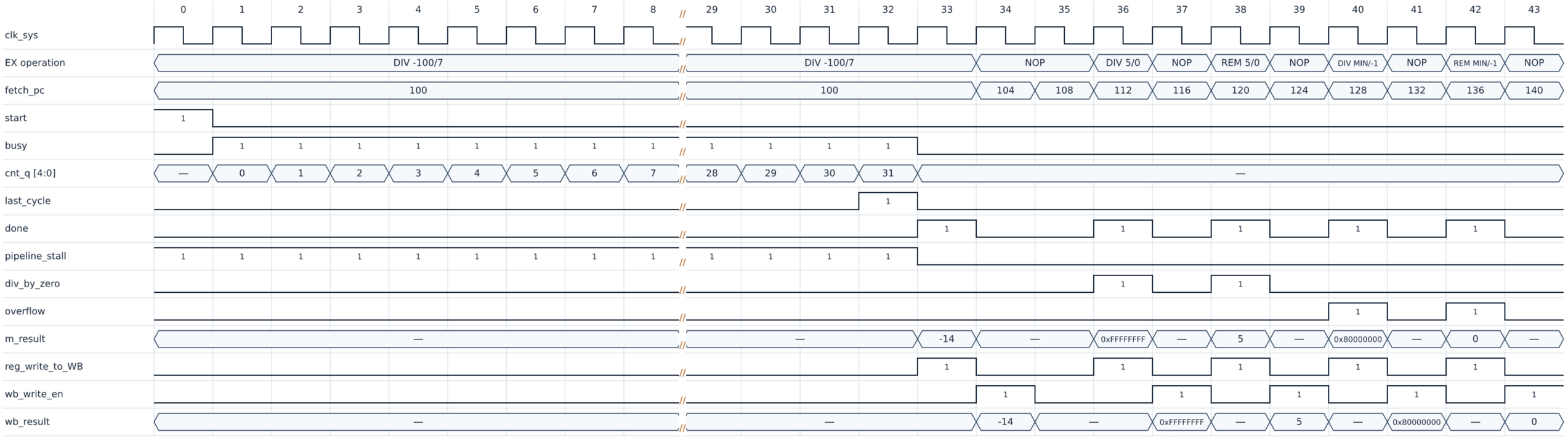

Initial test state: x10=0x100; x11=0x80000000; RAM[0x100]=32; P1_INPUT=0x10. Other consumed initial registers = 0.

H is BEQ at PC 28 -> PC 64. J/K must never commit. Independent illustrative sequence, not the real game software.

## B. RV32M — normal division and immediate special-case bypass

24 visible columns; clocks 9–28 OMITTED (20 genuine busy clocks). Normal DIV requires 32 iterations; real elapsed time exceeds 30.

Column = after left rising edge; outputs latch at next edge. — = not consumed. // = omitted time / independent boundary window.



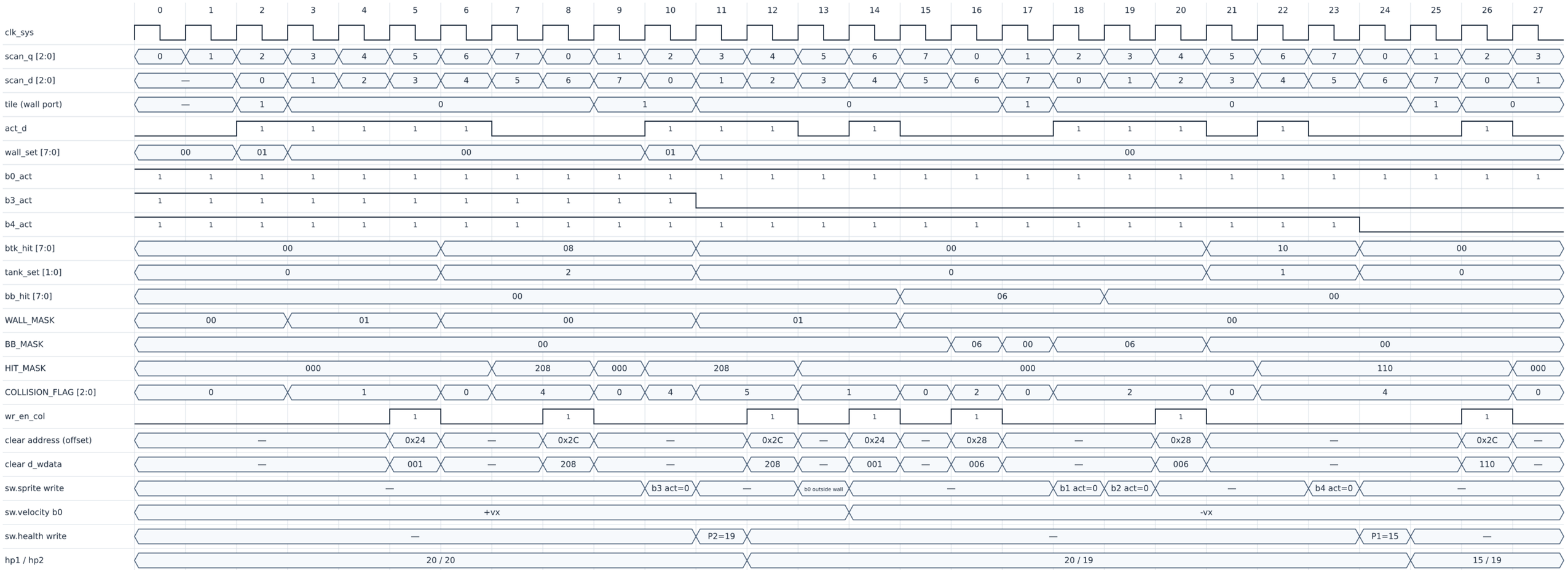

Start column 0; 32 iterations columns 1…32; done 33; WB 34. The // break omits clocks, not work.

DIVU/REMU have same handshake without signed correction. MUL high/low variants use funct3 without divider stall.

## C. Collision engine and software response — 28 continuous clocks

Preloaded bullets and maze; arbitrary test position updates. Wall scan, inactive suppression, normal/special tank hit, bullet pair, sticky clear priority.

Column = after left rising edge; outputs latch at next edge. — = not consumed. // = omitted time / independent boundary window.




All masks initially zero; maze and bullets already initialized. No reset is asserted in this independent continuous window.

Scan slot0 at 0 -> wall_set at 2 -> sticky mask at 3. After position write at 13, slot0 scan16 returns floor18.

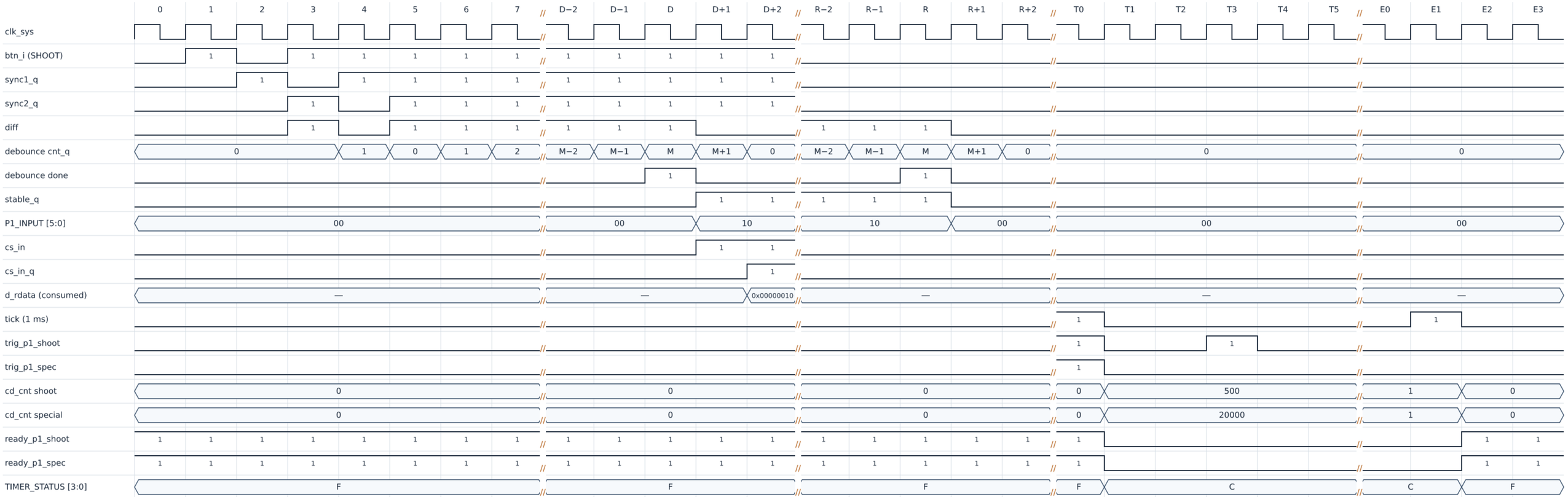
At 8/16 clear wins over a still-live hit; deactivation or separation must precede the final acknowledgement to avoid re-latching.

Travel/reflection inputs are test stimulus. No claim that a bullet traverses a maze, or software performs all these calculations, in 28 clocks.

## D. Button conditioning, cooldown trigger and expiry — 28 visible columns

Independent windows separated by //: debounce terminal D; release terminal R; timer trigger T; long cooldown expiry E. 50 MHz assumption.

Column = after left rising edge; outputs latch at next edge. — = not consumed. // = omitted time / independent boundary window.




Debounce gaps hide about 10 ms of sustained mismatch on each transition; M+1 column follows literal current-count comparator timing.

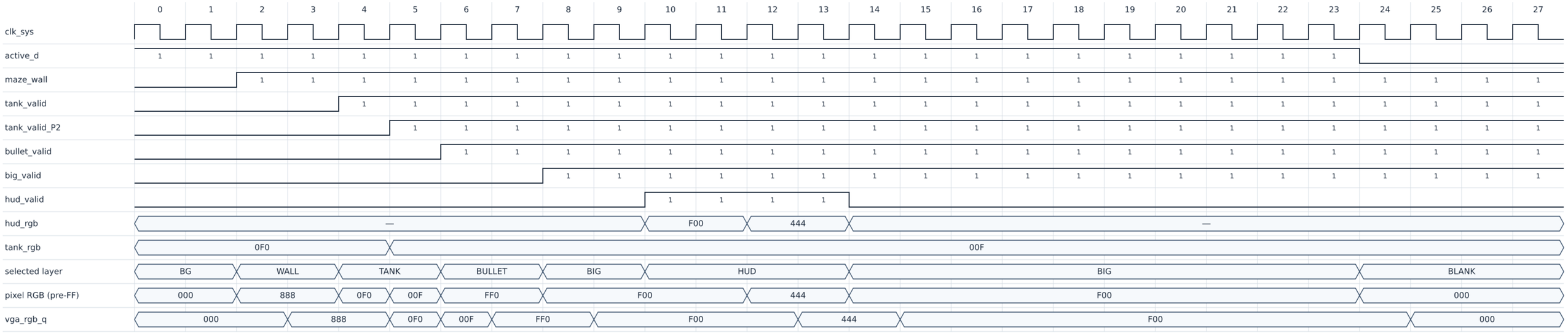
T0 coincides with a real terminal tick: trigger wins over decrement. E is an independent endpoint test with both counters initially 1; not a simultaneous expiry claim.

Normal software must check ready before shooting; hardware reloads even a busy channel if software writes a trigger. No FPGA fire-request pulse is drawn.

# E. Display pixel priority and blanking — 28 continuous clocks

Aligned Fig 10 input test: HUD > big bullet > normal bullet > tank > wall > background. Final RGB registered once.

Column = after left rising edge; outputs latch at next edge. — = not consumed. // = omitted time / independent boundary window.



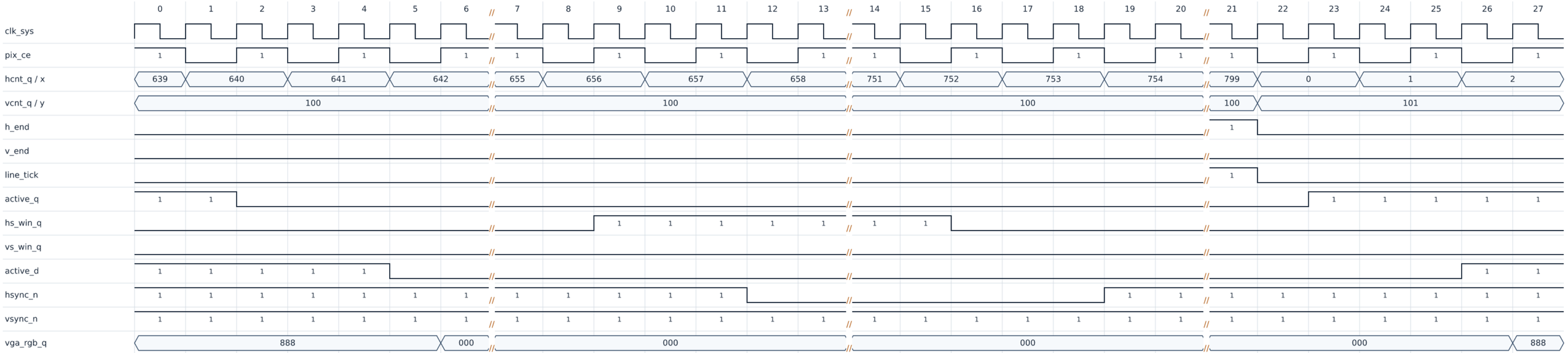

Overlapping valid streams prove priority without claiming a fast full-screen scan. This is a mixer-input test.

Transparent ROM pixels have valid=0 and reveal lower layers. Actual shape ROM bits are not specified in the block diagrams.

# F. VGA horizontal boundaries — visible blank, HS low, HS high, line wrap

Four independent seven-clock windows; // separates preconditioned boundary tests. 50 MHz clk_sys, alternating 25 MHz pix_ce.

Column = after left rising edge; outputs latch at next edge. — = not consumed. // = omitted time / independent boundary window.




Sync pin changes five displayed clocks after window start; RGB blank change follows one clock later under literal Fig 10.

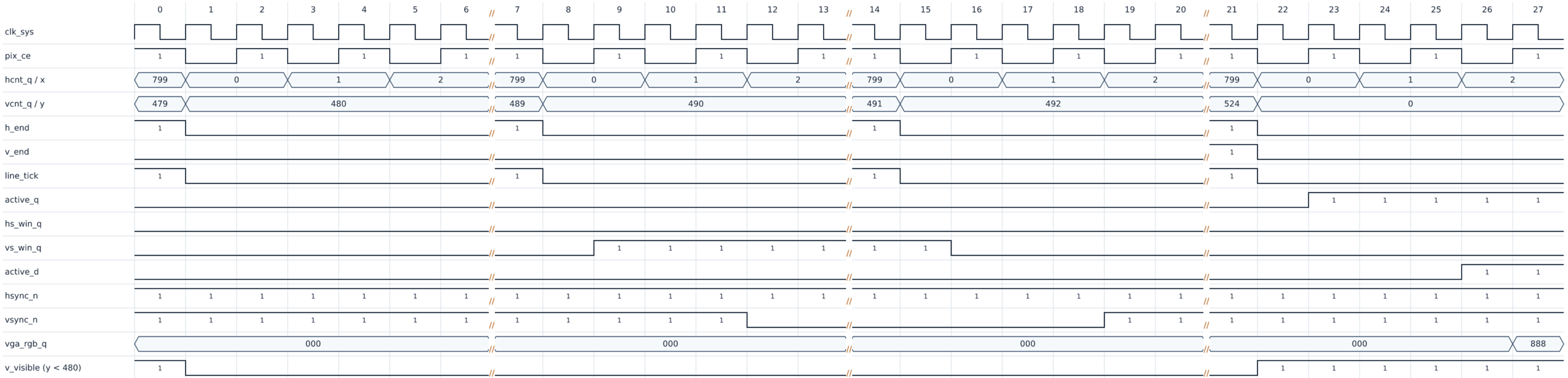
Each window has steady prehistory. // does not skip required counter values within a single continuous trace.

Exact coordinate-to-output alignment is NOT established by these conflicting drawings; verify/reset RTL before relying on pin timing.

# G. VGA vertical boundaries — vertical blank, VS low, VS high, frame wrap

Four independent seven-clock windows; // separates preconditioned boundary tests. 50 MHz clk_sys, alternating 25 MHz pix_ce.

Column = after left rising edge; outputs latch at next edge. — = not consumed. // = omitted time / independent boundary window.




Sync pin changes five displayed clocks after window start; RGB blank change follows one clock later under literal Fig 10.

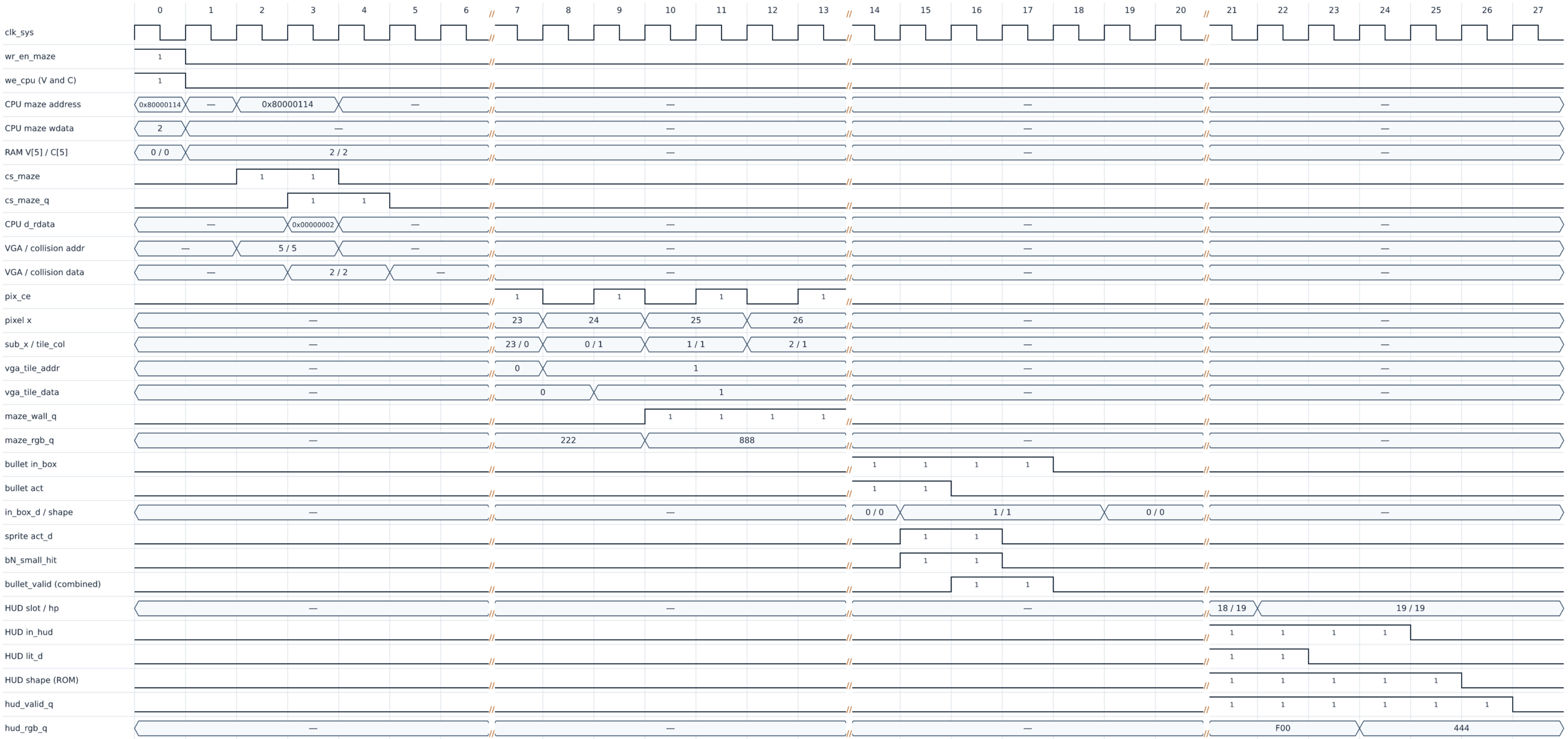
Each window has steady prehistory. // does not skip required counter values within a single continuous trace.

Exact coordinate-to-output alignment is NOT established by these conflicting drawings; verify/reset RTL before relying on pin timing.

# H. Maze coherence, tile boundary, sprite ROM and health HUD — 28 visible columns

Independent windows: mirrored maze write/read | 24-pixel tile crossing | bullet visibility | HUD heart valid/colour. No same-address read/write collision tested.

Column = after left rising edge; outputs latch at next edge. — = not consumed. // = omitted time / independent boundary window.




ROM outputs use illustrative shape bits since block diagrams contain no ROM image data. Streams are independent diagnostic tests.

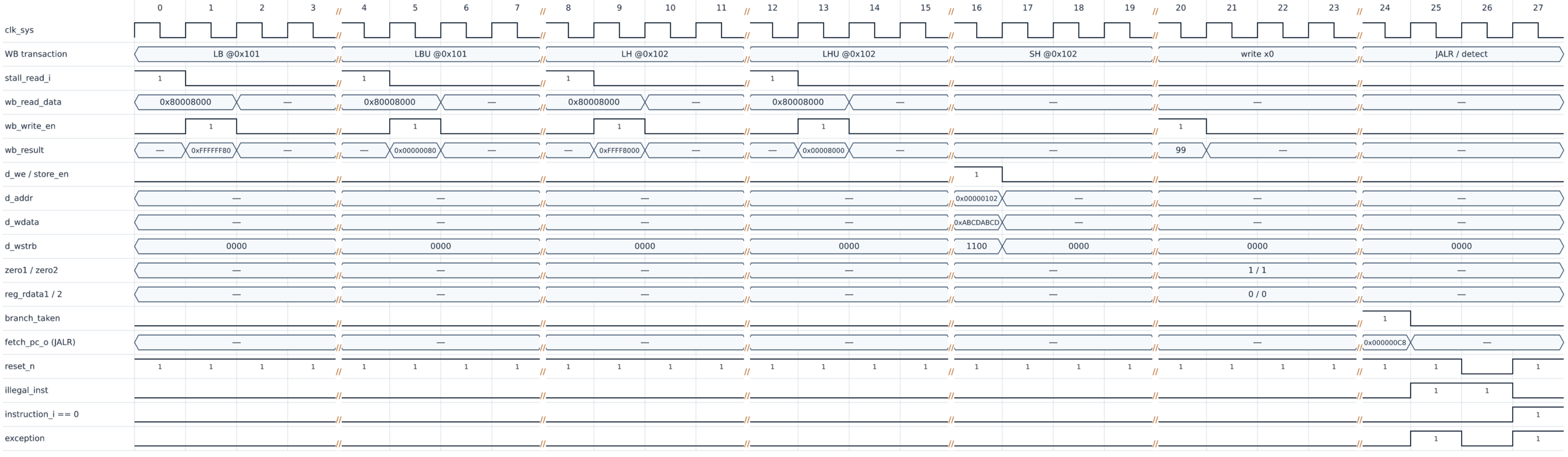
RAM same-address cross-port write/read mode and RAM reset contents are unspecified; this test avoids that collision and assumes preloaded tiles.

Unqualified X tile restart at h_end in Fig11 must be checked when pix_ce=0; boundary tests use a qualified restart convention.

# I. CPU corner cases — loads, halfword lanes, x0, jump and exception detect

Seven independent four-clock tests; // separates transactions. Valid inputs established before window. Trap/recovery not specified by diagrams.

Column = after left rising edge; outputs latch at next edge. — = not consumed. // = omitted time / independent boundary window.




Aligned data-memory tests only. MMIO is word-only; SB/SH MMIO behaviour should not be inferred from data-memory lane tests.

Normal ALU add/shift/logic/compare are selected combinationally; numeric examples in sheet A cover ADD, MUL, SLT, OR, LUI, SRAI and XOR.